

# The Effects of AI-Generated Summaries on Opinion Extremity in Online Discussions

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## Abstract

This project tests whether adding an AI-generated summary to an online discussion changes readers' opinions or their perceptions of public support. We conducted a randomized survey experiment with 230 U.S. participants. Each participant read three discussions, either with or without a summary, and reported their views before and after reading. Across the three topics, the estimated effect on opinion extremity was 0.01 points, with a 95% confidence interval from -0.15 to 0.17. We did not find evidence that the summaries reduced opinion extremity.

## Introduction

A summary can make a long discussion easier to read, but the choice of what to include may also change how readers interpret it. A reader could come away with a different view of the issue, or a different impression of how many other people support it. We study these two outcomes separately by adding a summary to otherwise identical Reddit-style discussions.

Prior research shows that AI-generated text can influence attitudes. Bai et al. [1] found that persuasive messages generated by language models changed policy attitudes. Tessler et al. [2] studied an AI mediator that developed statements of common ground through repeated input from participants. Xu et al. [3] found that search summaries shifted attitudes toward the summaries' framing. Our experiment tests a simpler intervention: one fixed AI-generated summary placed above the comments in an existing discussion.

We expected readers shown a summary to move closer to the midpoint of the opinion scale than readers shown no summary. We also tested whether summaries changed perceived public support, without specifying a direction for that change. We refer to distance from the midpoint as opinion extremity. This measures how strongly a reader supports or opposes a position, rather than whether the group is divided into opposing camps.

## Experiment

We recruited participants through Prolific and administered the survey in Qualtrics. Participants were based in the United States and completed the survey in English. The study was advertised as taking six minutes and paid \$1.58, equivalent to an advertised rate of \$15.80 per hour. The export contains 230 completed responses, with a median reported age of 39.

Participants first answered questions about familiarity, personal opinion, and perceived public support for each topic. Qualtrics then randomized assignment to a condition for all three discussions. The control group saw the original post and comments. The summary group saw the same material with an AI-generated summary between the post and the comments. Topic order was randomized within each condition. There were 114 participants in the control group and 116 in the summary group.

The topics were AI data center expansion, allowing police officers to mute or disable body cameras during incidents, and the death penalty for especially serious crimes. After each discussion, participants repeated the opinion and perceived-support questions and answered an attention check identifying the topic.

We collected the source discussions through Reddit's JSON output and used a Python script to convert the posts and comments into CSV files. The files were submitted to GPT-5.5 through ChatGPT using the prompt, "Can you produce an AI-generated summary for the following .csv file that was obtained from a reddit article?" We used one summary per topic, approximately 100 to 130 words long, and kept it fixed across participants. Summaries were labeled "AI-GENERATED SUMMARY." Usernames and vote counts were removed from both conditions.

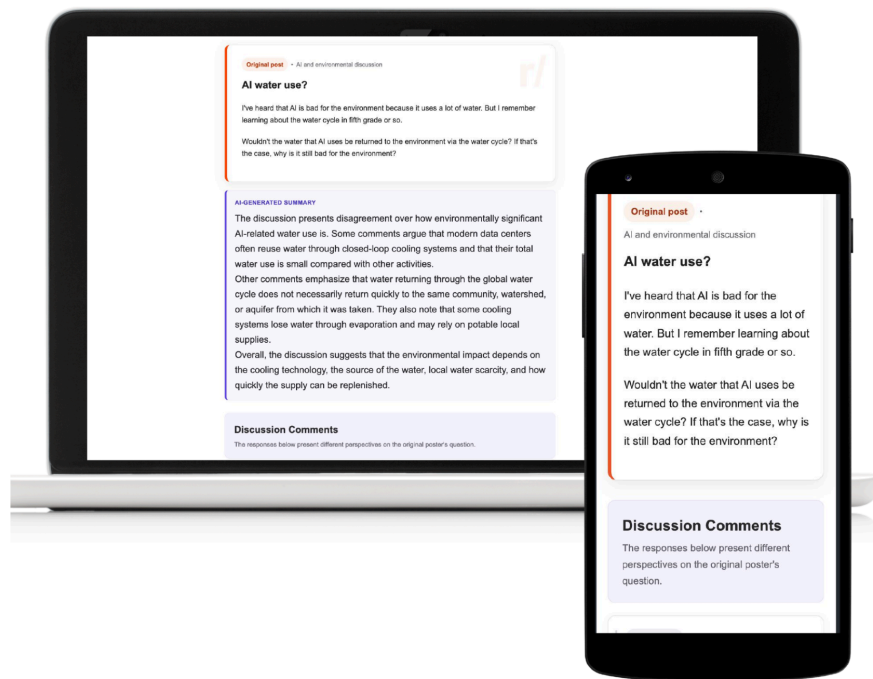


Figure 1 Example survey screens. The desktop view shows the AI-generated summary between the post and comments. The mobile view shows the post and comments without a visible summary. The control and summary conditions used identical comments, with usernames and vote counts removed.

## Data and measures

Each participant contributed one observation per topic, giving 690 observations. All 230 completed responses contained the opinion and perceived-support measures needed for the analysis. When preparing the analysis data, we identified condition from the populated control or summary question blocks and checked that each response contained data from only one condition. The shared dataset contains reassigned participant IDs and the response values needed for these analyses. It replaces exact reading times and survey durations with screening flags.

Personal opinion was measured from 1, strongly oppose, to 10, strongly support. We calculate opinion extremity as the absolute distance from 5.5. The outcome is post-reading extremity minus baseline extremity, so a negative change indicates movement toward the midpoint. Perceived public support asks how many people out of ten would support the stated position. Its outcome is also the post-reading value minus the baseline value.

For analysis, we exclude a topic observation if the participant answered its attention check incorrectly, spent less than ten seconds on the discussion, or completed the full survey in less than two minutes. The criteria overlap and exclude 25 observations: 20 in the control group and five in the summary group. The screened sample contains 665 observations from 226 participants. Because these measures are collected after assignment, screening may change the comparability of the groups. We therefore use it only as a sensitivity check. Figure 2 in the appendix shows the participant and observation counts.

## Analysis

We first compare mean changes between the two conditions across all three topics. We then fit adjusted linear models with topic fixed effects. For opinion extremity, the adjustment variable is baseline opinion on the original 1 to 10 scale. For perceived support, it is baseline perceived support. The treatment coefficient estimates the difference in change between the summary and control groups. In the unadjusted full sample, each participant contributes three observations and each topic receives equal weight.

We also test adjustment for baseline extremity, since strong opposition and strong support have the same distance from the midpoint. Baseline opinion about AI data centers was higher in the control group (4.32 versus 3.67). Adjustment allows the expected change to depend on the starting value; simply subtracting baseline does not guarantee that this difference is fully addressed. Appendix Table A1 reports baseline differences; Figure 3 shows the full-sample opinion distributions.

Standard errors are clustered by participant because each person contributes three observations. We use the CR1 finite-sample correction and a t distribution with the number of participants minus one degrees of freedom for 95% confidence intervals and two-sided p-values. We apply the same calculation to the topic-level estimates and sensitivity models. The reported p-values are not adjusted for multiple comparisons.

The initial power analysis focused on perceived consensus on a scale from zero to 100, while the final analysis emphasized opinion extremity. The planning simulation estimated roughly 80% power at 210 participants for an assumed eight-point consensus effect and an outcome standard deviation of 20. Those assumptions do not establish power for the extremity outcome used here. We report both outcomes and treat the comparisons as exploratory, rather than as preregistered tests.

## Results

### *Opinion extremity*

The pooled estimate is close to zero. In the full sample, the summary group's change in extremity was 0.011 points higher than the control group's change (95% confidence interval -0.153 to 0.174,  $p = .897$ ). Adjustment for baseline opinion and topic produced a similar estimate of 0.006 points. The screened sample also gave estimates close to zero. None of these models provides evidence that the summaries reduced opinion extremity.

Table 1 Estimated effects on opinion extremity

| Analysis            | Estimate | 95% confidence interval | p     | Observations (participants) |
|---------------------|----------|-------------------------|-------|-----------------------------|
| Full unadjusted     | 0.011    | -0.153 to 0.174         | 0.897 | 690 (230)                   |
| Full adjusted       | 0.006    | -0.159 to 0.171         | 0.944 | 690 (230)                   |
| Screened unadjusted | 0.011    | -0.157 to 0.179         | 0.895 | 665 (226)                   |
| Screened adjusted   | 0.006    | -0.164 to 0.176         | 0.946 | 665 (226)                   |

Estimates are summary minus control differences in change from baseline. Intervals use participant-clustered standard errors. Adjusted models include baseline opinion and topic fixed effects.

The separate topic estimates were -0.079 points for AI data centers, -0.010 for police body cameras, and 0.121 for capital punishment. All three confidence intervals included zero. Adjusting the pooled model for baseline extremity instead of signed baseline opinion changed the estimate to 0.091 points (95% confidence interval -0.073 to 0.254). This also did not show a reduction in extremity.

### *Perceived public support*

The unadjusted pooled estimate was -0.149 points on the scale from zero to ten (95% confidence interval -0.404 to 0.107,  $p = .252$ ). After adjustment for baseline perceived support and topic, the estimate was -0.231 points (95% confidence interval -0.456 to -0.005,  $p = .045$ ). The screened sample showed the same pattern: the unadjusted interval included zero, while the adjusted estimate was -0.240 points with an interval just below zero.

Table 2 Estimated effects on perceived public support

| Analysis            | Estimate | 95% confidence interval | p     | Observations (participants) |
|---------------------|----------|-------------------------|-------|-----------------------------|
| Full unadjusted     | -0.149   | -0.404 to 0.107         | 0.252 | 690 (230)                   |
| Full adjusted       | -0.231   | -0.456 to -0.005        | 0.045 | 690 (230)                   |
| Screened unadjusted | -0.171   | -0.435 to 0.093         | 0.202 | 665 (226)                   |
| Screened adjusted   | -0.240   | -0.469 to -0.012        | 0.040 | 665 (226)                   |

Estimates are summary minus control differences in change from baseline. Intervals use participant-clustered standard errors. Adjusted models include baseline perceived support and topic fixed effects.

These adjusted results suggest that readers shown a summary may have perceived less public support for the stated positions. However, the result depends on adjustment, and the full and screened samples contain mostly the same people. They are not independent confirmations. We examined several outcomes and specifications, so the p-values near .05 should be interpreted cautiously. Figure 4 plots the pooled estimates and their confidence intervals for both outcomes.

A stricter screen also excludes observations from participants who took more than 21 minutes, the platform's stated maximum completion time. It retains 634 observations from 215 participants. The adjusted estimates were 0.034 points for extremity (95% confidence interval -0.140 to 0.207) and -0.247 points for perceived support (95% confidence interval -0.484 to -0.010). This comparison has the same limitation as the other screened analysis: completion time is measured after treatment.

## Limitations

The treatment combines additional text, a particular summary, and an AI label. The experiment estimates the effect of that combination. It cannot separate the influence of the label from the content or the extra reading. We used only one summary for each of three discussions, so the results may depend on those specific summaries. We did not independently assess whether the summaries represented opposing views evenly.

Distance from the midpoint captures individual extremity, but it does not measure every form of polarization. A group can become more concentrated at one end of the scale without splitting into opposing camps. The survey also measures immediate self-reported responses, not whether changes persist or affect behavior.

The sample consists of U.S. Prolific respondents reading selected discussions in a survey. It may not represent people participating in live online discussions. The available export contains completed responses, so enrollment records are needed to assess whether anyone left after assignment. Finally, the estimates have uncertainty, and several related comparisons were examined. A follow-up study should specify its main outcome and model in advance, test multiple summaries per topic, and choose its sample size around an effect worth detecting.

## Conclusion

We did not detect a reduction in opinion extremity after participants read AI-generated summaries of three online discussions. Adjusted models estimated lower perceived public support in the summary group, but the evidence depended on the model specification. Whether summaries affect perceived public support more than readers' own opinions remains a question for a larger study.

## References

- [1] Bai, H., Voelkel, J. G., Muldowney, S., Eichstaedt, J. C., & Willer, R. (2025). LLM-generated messages can persuade humans on policy issues. *Nature Communications*, 16, 6037. <https://doi.org/10.1038/s41467-025-61345-5>
- [2] Tessler, M. H., Bakker, M. A., Jarrett, D., Sheahan, H., Chadwick, M. J., Koster, R., Evans, G., Campbell-Gillingham, L., Collins, T., Parkes, D. C., Botvinick, M., & Summerfield, C. (2024). AI can help humans find common ground in democratic deliberation. *Science*, 386(6719), eadq2852. <https://doi.org/10.1126/science.adq2852>
- [3] Xu, Y., Dash, S., Kang, S., Liao, W., & Spiro, E. S. (2025). AI summaries in online search influence users' attitudes. *arXiv:2511.22809v2*. <https://doi.org/10.48550/arXiv.2511.22809>

## Appendix: Sample and baseline checks

Participant and observation flow

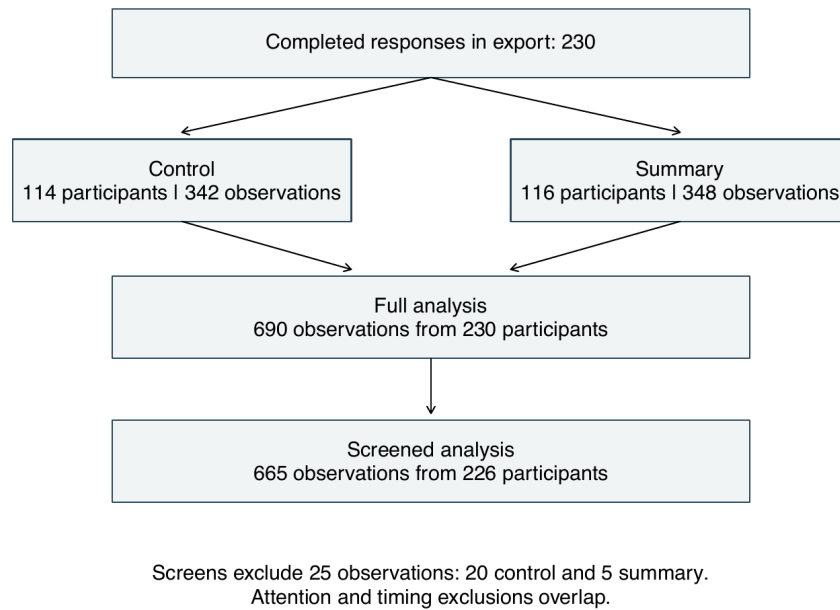


Figure 2 Participant and observation flow. The diagram starts with completed responses in the available export. It does not establish whether anyone left the survey after assignment. Each observation is one participant-topic response.

Table A1 Baseline covariate balance

| Measure                                  | Control | Summary | Difference | Std. diff. | p     |
|------------------------------------------|---------|---------|------------|------------|-------|
| Age as reported                          | 42.48   | 42.24   | -0.24      | -0.02      | 0.910 |
| AI data centers<br>Opinion               | 4.32    | 3.67    | -0.65      | -0.27      | 0.044 |
| AI data centers<br>Perceived support     | 4.18    | 3.78    | -0.39      | -0.20      | 0.127 |
| Police body cameras<br>Opinion           | 2.74    | 2.40    | -0.34      | -0.14      | 0.302 |
| Police body cameras<br>Perceived support | 3.42    | 3.16    | -0.26      | -0.12      | 0.357 |
| Capital punishment<br>Opinion            | 5.77    | 5.47    | -0.31      | -0.10      | 0.440 |
| Capital punishment<br>Perceived support  | 5.63    | 5.72    | 0.09       | 0.06       | 0.668 |

Control n = 114; summary n = 116. Differences are summary minus control; standardized differences divide by the pooled within-group standard deviation. P-values use two-sided Welch tests and are descriptive. Age includes the reported value of 120. A large p-value does not establish balance.

## Appendix: Baseline distributions and pooled effects

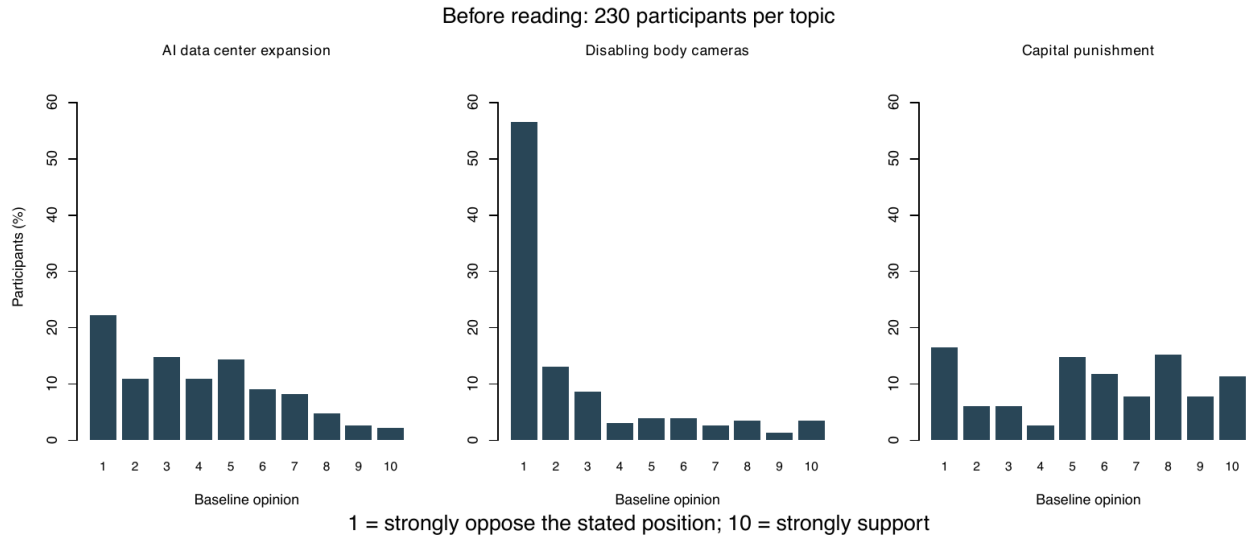


Figure 3 Baseline opinion distributions by topic. All 230 participants are included before screening. The police question concerns allowing officers to mute or disable body cameras during incidents. Concentration near one end of the scale does not by itself indicate two opposing groups.

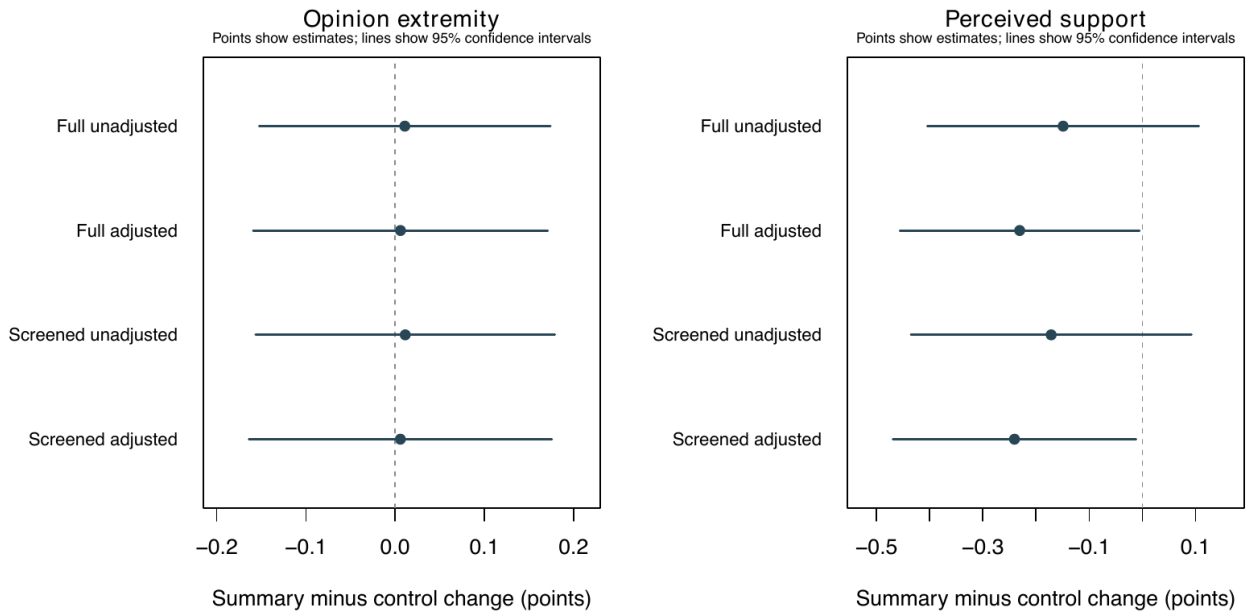


Figure 4 Pooled treatment estimates with 95% confidence intervals. Points are summary minus control differences in change; intervals use participant-clustered CR1 standard errors. The full sample has 690 observations from 230 participants; the screened sample has 665 observations from 226 participants. Tables 1 and 2 report the exact estimates and adjustment variables.